

Local degree conditions for the size Ramsey numbers of star forests

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Abstract

For two star forests, the Burr–Erdős–Faudree–Rousseau–Schelp conjecture expresses their size Ramsey number as a sum of antidiagonal maxima of component sizes. We prove a sufficient condition combining the parity of maximizing components with local drops between consecutive antidiagonal maxima. The condition permits arbitrarily many components and long intervals of equal maxima. The simultaneous-suffix framework and parity-dependent lower bound were previously recorded by rickyc on the Erdős Problems forum. Our additional step controls the maximum degrees of equality hosts to recover the conjectured value under local gap conditions. As a concrete consequence, for every $r, m \geq 1$,

$$\widehat{r}(K_{1,6} \sqcup rK_{1,3} \sqcup mK_2, K_{1,5} \sqcup K_2) = 11 + 7r + 5m.$$

We also obtain the conjectured formula whenever consecutive antidiagonal maxima decrease by at least two. These results do not settle the general conjecture.

Draft status. AI assistance was used in developing and writing this draft. The coordinating human reports having mathematically reviewed it; external peer review has not taken place. Priority of the additional gap criterion remains to be confirmed. This is a partial result, not a claim that the general Erdős problem is solved.

1 Statement

All graphs are finite and simple. Copies need not be induced. Write $G \rightarrow (F, H)$ if every red/blue coloring of $E(G)$ contains a red copy of F or a blue copy of H . The size Ramsey number $\widehat{r}(F, H)$ is the minimum of $e(G)$ over these hosts.

Let $A = (a_1, \dots, a_s)$ and $B = (b_1, \dots, b_t)$ be nonincreasing sequences of positive integers, and put

$$F(A) = \bigsqcup_{i=1}^s K_{1,a_i}, \quad F(B) = \bigsqcup_{j=1}^t K_{1,b_j}, \quad q = s + t.$$

For $2 \leq k \leq q$, define

$$\ell_k = \max_{\substack{1 \leq i \leq s, 1 \leq j \leq t \\ i+j=k}} (a_i + b_j - 1), \quad S_k = \sum_{h=k}^q \ell_h.$$

The conjecture of Burr et al. [1] asserts $\widehat{r}(F(A), F(B)) = S_2$.

Call an index k *good* if at least one pair (i, j) attaining ℓ_k has either both a_i, b_j odd, or $\min(a_i, b_j) = 1$. The word good refers only to this explicit arithmetic condition.

Theorem 1 (Local criterion). *Suppose that every index $2 \leq k < q$ which is not good satisfies at least one of the following conditions:*

1. $\ell_k - \ell_{k+1} \geq 3$;
2. $\ell_k - \ell_{k+1} = 2$, and either

$$k+1 < q \text{ and } \ell_{k+1} - \ell_{k+2} \geq 2, \quad \text{or} \quad k+1 = q \text{ and } \ell_q \geq 2.$$

There is no condition on the final index q . Then

$$\hat{r}(F(A), F(B)) = S_2.$$

Every host attaining this value has maximum degree at most ℓ_2 . If $q > 2$ and $\ell_2 - \ell_3 \geq 2$, it has at most one vertex of degree ℓ_2 . For $q = 2$, the last assertion holds when $\ell_2 \geq 2$.

The all-odd hypothesis in Davoodi et al. [3] makes every index good. Theorem 1 also allows mixed parity and equal successive values of ℓ_k at good indices. Its conditions involve the maximizing pairs and adjacent values, not the total number of components. Single-star cases, including a star against two arbitrary positive stars, also appear in Lortz and Mengersen [7, Theorem 7].

A forum contribution by rickyc dated 6 August 2026 [8] already gives the simultaneous-suffix deletion framework, the same distinction between good and other diagonals, and the general lower bound

$$\hat{r}(F(A), F(B)) \geq S_2 - \sum_{k=2}^{q-1} \eta_k, \quad \eta_k = \begin{cases} 0, & k \text{ is good,} \\ 1, & k \text{ is not good.} \end{cases}$$

We do not claim these ingredients as new. The candidate addition here is the equality-host degree argument in Section 3: under the local gap conditions it removes all of these deficits. The August contribution was identified during the final attribution audit, after the present argument had been developed.

2 Deletion and degree bounds

For $1 \leq i \leq s$ and $1 \leq j \leq t$, let

$$A_i = (a_i, \dots, a_s), \quad B_j = (b_j, \dots, b_t).$$

A graph has property $\mathcal{P}_k$ if

$$G \rightarrow (F(A_i), F(B_j)) \quad \text{for every feasible } i + j = k.$$

In particular, $\mathcal{P}_2$ is the original Ramsey property. Property $\mathcal{P}_q$ concerns the two single stars K_{1,a_s} and K_{1,b_t} .

Lemma 2 (Simultaneous suffix deletion). *If G has $\mathcal{P}_k$ and $k < q$, then $G - v$ has $\mathcal{P}_{k+1}$ for every vertex v .*

This formulation and its proof already appear in [8]; we include the argument to make the presentation self-contained.

Proof. First suppose $G \rightarrow (F(A_i), F(B_j))$ and $i < s$. Suppose an edge coloring of $G - v$ avoids red $F(A_{i+1})$ and blue $F(B_j)$. Extend it by coloring all edges at v red. A red $F(A_i)$ in the extension would have at most one component meeting v . The other components contain $F(A_{i+1})$, since removing any one star from a nonincreasing list leaves a forest containing the list with its largest star removed. A blue target is unchanged. Thus no such avoiding coloring exists. The analogous assertion holds on deleting the largest blue star.

For each feasible pair (i, j) with $i + j = k + 1$, at least one of $(i - 1, j)$ and $(i, j - 1)$ is feasible. Apply the corresponding assertion to that predecessor. $\square$

We use Vizing's theorem and the following result of Fournier: if the subgraph induced by the maximum-degree vertices of a graph is a forest, the graph has a proper edge coloring with $\Delta(G)$ colors; see [4]. We call this induced subgraph the maximum-degree core.

Lemma 3. *If G has $\mathcal{P}_k$, then $\Delta(G) \geq \ell_k - 1$. If k is good, then $\Delta(G) \geq \ell_k$.*

Proof. Choose a pair (i, j) attaining ℓ_k , and write $a = a_i$, $b = b_j$. If $\Delta(G) \leq a + b - 3$, Vizing gives at most $a + b - 2$ matching color classes. Merge at most $a - 1$ classes into red and at most $b - 1$ into blue. This avoids both largest stars and therefore avoids both target forests, a contradiction.

If a, b are odd, a graph of maximum degree at most $a + b - 2$ also admits a coloring with red maximum degree at most $a - 1$ and blue maximum degree at most $b - 1$. Indeed, embed it in an $(a + b - 2)$ -regular simple graph, factor that even-regular graph into 2-factors, and put $(a - 1)/2$ factors in red and $(b - 1)/2$ in blue. Restrict to the original graph. The zero-degree case is immediate. This is the parity argument in [3, Lemma 2.1]. A regular supergraph can be obtained by repeatedly taking two disjoint copies and joining corresponding deficient vertices.

If $a = 1$, the all-blue coloring requires a copy of the blue forest, hence degree at least $b = \ell_k$. The case $b = 1$ is symmetric. Applying either argument to a witnessing maximizing pair proves the last assertion. $\square$

Lemma 4 (Upper construction). *The disjoint union $\bigsqcup_{h=k}^q K_{1, \ell_h}$ has $\mathcal{P}_k$.*

Proof. Fix a feasible starting pair (i, j) with $i + j = k$. At the star with ℓ_{i+j} edges, the center has at least a_i red edges or at least b_j blue edges. Select the corresponding monochromatic star, and increase i or j , respectively. Continue on the next host component. One list is exhausted in at most $q - k + 1$ steps. The selected stars in that color form its target forest, and are vertex-disjoint because they came from different host components. $\square$

3 Proof of the local criterion

Proof of Theorem 1. We prove, by decreasing induction on k , three assertions:

1. every graph with $\mathcal{P}_k$ has at least S_k edges;
2. every such graph with exactly S_k edges has maximum degree at most ℓ_k ;
3. an equality graph has at most one degree- ℓ_k vertex if $k < q$ and $\ell_k - \ell_{k+1} \geq 2$, or if $k = q$ and $\ell_q \geq 2$.

For $k = q$ this is the elementary formula

$$\widehat{r}(K_{1, a_s}, K_{1, b_t}) = a_s + b_t - 1 = \ell_q.$$

For completeness, any graph with at most $a_s + b_t - 2$ edges can be colored with at most $a_s - 1$ red edges and at most $b_t - 1$ blue edges. The opposite inequality follows from a star with $a_s + b_t - 1$ edges. An equality graph has maximum degree at most its edge count. If two vertices had degree $\ell_q \geq 2$, their incident edges would number at least $2\ell_q - 1 > \ell_q$, which is impossible.

Now let $k < q$ and assume the assertions for all larger indices. Take a graph G with $\mathcal{P}_k$, a maximum-degree vertex v , and set $D = \Delta(G)$, $H = G - v$. By Lemma 2, H has $\mathcal{P}_{k+1}$, so

$$e(G) = D + e(H) \geq D + S_{k+1}.$$

If k is good, Lemma 3 gives $D \geq \ell_k$ and the lower bound follows. Otherwise suppose, for a contradiction, that $e(G) < S_k$. Lemma 3 and integrality force

$$e(G) = S_k - 1, \quad D = \ell_k - 1, \quad e(H) = S_{k+1}. \quad (1)$$

The inductive equality bound gives $\Delta(H) \leq \ell_{k+1}$.

If $\ell_k - \ell_{k+1} \geq 3$, every vertex other than v has degree in G at most $\ell_{k+1} + 1 \leq D - 1$. The maximum-degree core consists of v alone.

In the remaining allowed case, $\ell_k - \ell_{k+1} = 2$. Every degree- D vertex other than v has degree exactly $\ell_{k+1} = D - 1$ in H and is adjacent to v . By the stated additional condition and the inductive third assertion, there is at most one such vertex. Consequently the maximum-degree core of G is a vertex or an edge.

In either case Fournier's theorem yields a proper edge coloring with D matching classes. For a pair (i, j) attaining ℓ_k , equation (1) says $D = a_i + b_j - 2$. Merge $a_i - 1$ classes into red and $b_j - 1$ into blue. Neither color contains its largest target star. This contradicts $\mathcal{P}_k$, proving the lower bound.

It remains to prove the equality assertions at index k . If $e(E) = S_k$ and E has $\mathcal{P}_k$, deleting any vertex w and applying the lower bound at $k + 1$ gives

$$d_E(w) \leq S_k - S_{k+1} = \ell_k.$$

Suppose $\ell_k - \ell_{k+1} \geq 2$ and w, u are distinct degree- ℓ_k vertices. Then $E - w$ has $\mathcal{P}_{k+1}$ and exactly S_{k+1} edges, so its maximum degree is at most ℓ_{k+1} . But u has degree at least $\ell_k - 1 > \ell_{k+1}$ there, a contradiction. This completes the induction. Lemma 4 supplies the matching upper bound. $\square$

4 Consequences and comparison

Corollary 5 (Decreasing maxima). *If $\ell_k - \ell_{k+1} \geq 2$ for every $2 \leq k < q$, the conjectured formula holds.*

Proof. If $\ell_q \geq 2$, every nonterminal index satisfies one of the local conditions. If $\ell_q = 1$, then $a_s = b_t = 1$. Every feasible pair on the penultimate diagonal has one component of size 1, so $q - 1$ is good. Every earlier index again satisfies a local condition. For $q = 2$, the single-star formula applies directly. $\square$

Corollary 6 (An even largest component). *Suppose $s, t \geq 2$, all components except a_1 have odd size, and a_1 is even. If*

$$a_1 - a_2 \geq 3, \quad b_1 - b_2 > a_1 - a_2,$$

then the conjectured formula holds provided, for every $3 \leq k \leq \min(q, t + 1)$, some maximizing pair on diagonal k has $i \geq 2$.

Proof. The additional maximizing-pair condition makes every index after the first good; for $k > t + 1$, row $i = 1$ is unavailable. On diagonal 3, the strict gap inequality yields

$$\ell_3 = a_2 + b_1 - 1, \quad \ell_2 - \ell_3 = a_1 - a_2 \geq 3.$$

Thus the only possibly bad nonterminal index satisfies the first local condition. $\square$

For $t = 2$, the extra maximizing-pair condition in Corollary 6 is automatic from its strict gap inequality. This already gives unrestricted lengths and repetitions in the tail of A .

Corollary 7 (A family with two independent multiplicities). *For all positive integers r, m ,*

$$\widehat{r}(K_{1,6} \sqcup rK_{1,3} \sqcup mK_2, K_{1,5} \sqcup K_2) = 11 + 7r + 5m.$$

Proof. The antidiagonal maxima, in order, are

$$10, \underbrace{7, \dots, 7}_{r \text{ times}}, \underbrace{5, \dots, 5}_{m \text{ times}}, 1.$$

The first index is not good and is followed by a drop of 3. Every later maximum is attained by an odd-odd pair. Theorem 1 gives the sum displayed in the statement. $\square$

Another example, with both components of the second forest larger than one, is

$$\widehat{r}(K_{1,6} \sqcup K_{1,3} \sqcup rK_2, K_{1,7} \sqcup K_{1,3}) = 24 + 7r \quad (r \geq 1).$$

Here the maxima are 12, 9, followed by r copies of 7, and finally 3. Again only the first index is bad, with a drop of 3.

Corollary 8 (Arbitrarily many even components). *Let $h, m, r_1, \dots, r_h$ be positive integers. Construct A by concatenating the blocks*

$$\left(4j + 2, \underbrace{4j - 1, \dots, 4j - 1}_{r_j \text{ entries}} \right) \quad \text{for } j = h, h - 1, \dots, 1,$$

and then appending m entries equal to 1. For $B = (5, 1)$,

$$\widehat{r}(F(A), F(B)) = 2h^2 + 8h + \sum_{j=1}^h r_j(4j + 3) + 5m + 1.$$

This value exceeds the lower bound obtained by rounding every even component of A down by one and applying the all-odd theorem by exactly h .

Proof. Consecutive entries of A differ by at most 3. With $B = (5, 1)$, every diagonal except the last is therefore maximized by its column-1 pair: the competing column-2 value is the preceding entry of A , whereas the column-1 value is the current entry plus 4. The maxima are the entries of A plus 4, followed by 1.

The only bad indices correspond to the h even entries $4j + 2$. Each is followed by $4j - 1$, so its maximum drops by exactly 3. All other indices are good. Summing the maxima gives

$$\sum_{j=1}^h (4j + 6) + \sum_{j=1}^h r_j(4j + 3) + 5m + 1,$$

which is the claimed expression. If each even entry is reduced by one, the resulting sequence is still nonincreasing and all its adjacent gaps are at most 2. Column 1 still supplies every nonfinal maximum; exactly h of those maxima have decreased by one. Both forests now have only odd components, so their size Ramsey number is the corresponding sum by the published all-odd theorem. Monotonicity therefore gives precisely the claimed comparison lower bound. $\square$

What the comparison establishes. The families in Corollaries 7 and 8 have neither forest uniform, contain an even component, and have more than one nontrivial component in the first forest. They therefore fall outside the uniform, all-odd, one-star, and equal-two-star hypotheses listed in [3], as well as the one-nontrivial-star-per-forest scope of Cheng’s thesis [5]. Their final maximum is 1, so they fail the Győri–Schelp numerical sufficient condition as restated in [3, Theorem 1.4]. For the example with $B = (7, 3)$, that numerical condition fails already at the first 7 when $r \geq 3$, because $\binom{7}{2} = 21 < 7r + 3$.

For Corollary 7, the bound in [8] is one edge below the exact value proved here. For Corollary 8, it is h edges below, since exactly h nonfinal diagonals are not good. Thus the growing improvement also holds relative to that specific prior forum bound, not only to the direct all-odd rounding comparison.

These observations establish noncoverage by those stated hypotheses, not priority over all results or consequences in the literature. The full original Győri–Schelp article and the full Cheng thesis were not accessible in this audit. The proof of Theorem 1 does not depend on either source. A separate novelty audit records the accessible papers and the limitations of the search.

What remains open. The theorem does not cover a bad diagonal followed by a plateau, or a drop of one, unless another theorem handles that instance. For example, many forests containing long runs of even star sizes remain outside the criterion. No resolution of the full star-forest conjecture is asserted.

References

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The original version’s general-solution claim was removed in later versions.

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